Bob went to the store. Bob gave a bottle to", we display the identity scores computed by comparing attention head outputs against the answer token direct output difference. The bottom row shows head importance from EAP-IG-inputs for comparison. Both methods identify the same critical heads in layers 9–11. Interestingly, this simple analysis also reveals that the token "gave" exhibits a similar activation structure to the final token "to", suggesting that the model had as an option to output "Mary" after it. This illustrates how answer token direction encodes their own generation.

4.3. Edge-Level Circuit Discovery

Node-level scores (Equation 5) capture a component's *direct* importance—its contribution to the final output. However, a component's *total* importance includes its influence on downstream components. A head may have low direct importance but be critical because it shapes the queries or keys of a later head that does the heavy lifting. Similarly, an MLP may matter primarily through how downstream components read its output.

To capture this, we analyze edges: the pathways through which one component influences another. The residual stream at any layer is the sum of all upstream component outputs. We can therefore decompose a downstream component's task-relevant input into contributions from each upstream source.

For a target attention head j, information arrives through three channels (Q, K, V). Consider the key channel: the answer token difference $\Delta\mathbf{k}^{(j)}$ in key space can be projected back to residual stream space via $W_K^{(j)}\Delta\mathbf{k}^{(j)}$. This is the direction in residual stream space that carries task-relevant information for j's key computation.

The fraction of this signal contributed by upstream component i is:

$$R_{i\to j}^{(K)} = \frac{\left\langle \Delta\mathbf{o}_i,\ W_K^{(j)}\Delta\mathbf{k}^{(j)} \right\rangle}{\left\langle \Delta\mathbf{r}^{(\ell_j)},\ W_K^{(j)}\Delta\mathbf{k}^{(j)} \right\rangle} \tag{6}$$

where $\Delta\mathbf{o}_i$ is the differential output of component i and $\Delta\mathbf{r}^{(\ell_j)}$ is the full residual stream difference at j's layer. By linearity, $\sum_i R_{i\to j}^{(K)} = 1$: the per-channel ratios sum to one across all upstream components. Analogous expressions give $R_{i\to j}^{(Q)}$ and $R_{i\to j}^{(V)}$.

For edges targeting MLPs, there is only one pathway through the residual stream, so we compute a single ratio using the MLP's input-space target.

4.4. Channel Attribution via Shapley Decomposition

Equation 6 scores edges through each channel independently, but how should we weight Q, K, and V when aggregating a head's total edge importance? Rather than choosing weights arbitrarily, we derive them from a principled decomposition of the head's contribution.

We frame this as a cooperative game with three players. Each player (Q, K, V) can either carry task-relevant information from the base prompt (clean) or the contrastive prompt (corrupted); the question is how to divide credit among them based on their contribution to the head's total importance. Shapley values provide a principled answer: each player's contribution is its average marginal effect across all possible orderings of players (Shapley, 1953). For n players, the Shapley value for player i is:

$$\phi_i = \sum_{\mathcal{C}\subseteq N\setminus\{i\}} \frac{|\mathcal{C}|!(n-|\mathcal{C}|-1)!}{n!}\left[v(\mathcal{C}\cup\{i\}) - v(\mathcal{C})\right] \tag{7}$$

where $v(\mathcal{C})$ is the value when only players in $\mathcal{C}$ are active.

For our three-player game, this expands to a closed form. Taking ϕ_Q as an example:

$$\phi_Q = \frac{1}{3}\left[S_Q - S_\emptyset\right] + \frac{1}{6}\left[S_{QK} - S_K\right] + \frac{1}{6}\left[S_{QV} - S_V\right] + \frac{1}{3}\left[S_{QKV} - S_{KV}\right] \tag{8}$$

where $S_\mathcal{C}$ denotes the head's importance when only channels in $\mathcal{C}$ use clean activations. Analogous expressions hold for ϕ_K and ϕ_V.

Measuring coalition values. To compute $S_\mathcal{C}$, we run the head with a mixture of clean and corrupted activations: channels in $\mathcal{C}$ use activations from the clean prompt, while channels in $\{Q, K, V\} \setminus \mathcal{C}$ use activations from the corrupted prompt. We then measure the head's output alignment with the target direction (Equation 5) under this configuration. From Eq.7 this requires evaluating $2^3 = 8$ coalitions per head.

Additivity. The Shapley decomposition preserves the additivity property: $\phi_Q + \phi_K + \phi_V = S_{QKV} - S_\emptyset$, ensuring that channel contributions sum to the head's total importance.

Combining edge ratios and channel weights. The total edge importance from component i to attention head j combines the per-channel ratios (Equation 6), the Shapley weights, and the target component's total importance:

$$E_{i\to j} = S_j \cdot \left(\phi_Q^{(j)} \cdot R_{i\to j}^{(Q)} + \phi_K^{(j)} \cdot R_{i\to j}^{(K)} + \phi_V^{(j)} \cdot R_{i\to j}^{(V)}\right) \tag{9}$$

where S_j is the total importance of head j from Equation 5. This gives the amount of task-relevant signal flowing from component i to head j.

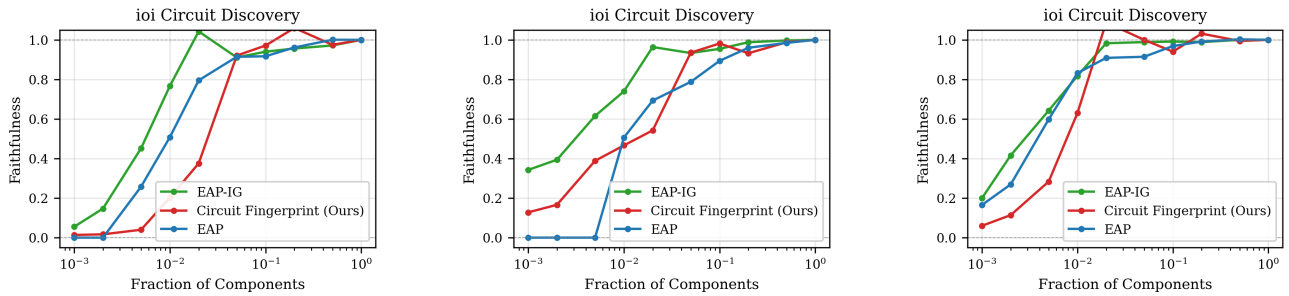

Figure 3. Edge-level circuit discovery on IOI. **(Left)** GPT2-Small. **(Center)** Qwen2.5-0.5B. **(Right)** Llama3.2-1B.

Algorithm 1 Computing Total Component Importance

1. Compute direct importance S_c for all components
2. Compute Shapley ratios $\phi_Q^{(j)}, \phi_K^{(j)}, \phi_V^{(j)}$ for all heads
3. Initialize total importance: $T_c \leftarrow S_c$
4. **for** $\ell = L$ down to 1 **do**
   for each component j at layer ℓ **do**
   for each upstream component i **do**
   Compute $E_{i\to j}$ using T_j (Eq. 9)
   $T_i \leftarrow T_i + E_{i\to j} \cdot T_j$
5. **return** T_c for all components

For edges targeting MLPs, there is no Shapley decomposition, so:

$$E_{i\to m} = S_m \cdot R_{i\to m} \tag{10}$$

Total component importance. A component's total importance is the sum of its direct importance and its indirect importance through all downstream edges. Since edges depend on the target component's total importance, we compute this via a backward pass through the model.

This propagates importance backward: components in later layers have their total importance computed first, which is then used to compute the edge contributions to earlier components. Algorithm 1 presents our edge attribution procedure. We emphasize that our goal is not to propose an optimal circuit discovery method, but to demonstrate that geometric alignment with answer tokens recovers circuit structure. To simplify the procedure, we focus on the final token position, ignoring indirect effects from earlier positions and the influence of layer normalization. The algorithm serves as a vehicle for validating the geometric property; the contribution is the relationship itself, not the particular implementation.

5. Writing: Geometric Steering

If circuit fingerprints genuinely encode computational structure, the same geometric space used for circuit discovery should enable controlled steering. This provides causal validation: directions that *read* circuit structure should also *write* predictable behavioral changes. If circuits follow a predictable space we can use these spaces to steer them.

Method. We construct an intervention subspace from answer token representations semantically related to the source and target features—where the source is the feature present in the current context and the target is the feature we wish to inject into the response. We intervene at attention heads identified by our circuit discovery method, operating in the same head-dimensional space where circuit fingerprints are computed. Given answer token representations $\{r_1, \ldots, r_k\}$, we center by subtracting the mean μ and compute an orthonormal basis $\{u_1, \ldots, u_m\}$ via SVD. The source and target directions are obtained by projecting the centered answer prototypes onto this basis:

$$d_s = \sum_{i=1}^{m} ((\bar{r}_s - \mu) \cdot u_i)\, u_i \tag{11}$$

where $\bar{r}_s$ is the source answer prototype (analogously for d_t).

When the target token is known (e.g., factual recall tasks), the steering intervention uses the separation between source and target as the intervention scale:

$$X' = X - \|d_s - d_t\|\hat{d}_s + \|d_s - d_t\|\hat{d}_t \tag{12}$$

where $\hat{d}_s$ and $\hat{d}_t$ are unit vectors.

For stylistic features (e.g., emotion, language) where the target token is not known at intervention time, we transfer the source magnitude instead:

$$X' = X - \|d_s\| \left(\hat{d}_s - \hat{d}_t\right) \tag{13}$$

This equation allows us to steer text generation towards different styles.

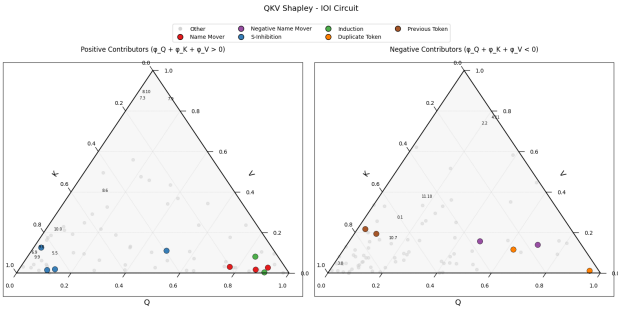

Figure 4. **Circuit function predicts QKV decomposition.** Shapley values decompose each head's contribution into Query, Key, and Value components. Heads with similar circuit roles cluster together: output-focused heads (Name Movers) are Q-dominated, while routing heads (S-Inhibition) are K-dominated.

Table 1. Circuit discovery results. Our method Circuit Fingerprint (CF) achieves comparable CMD and CPR to gradient-based baselines (EAP, EAP-IG) across three tasks and four model families. MCQA is not used in GPT2-Small because there is no circuit in the model for it.

Model	Method	IOI		SVA		MCQA	
		CMD	CPR	CMD	CPR	CMD	CPR
GPT2-Small	EAP	0.03	0.97	0.06	0.94	N/A	N/A
	EAP-IG-inputs	0.03	0.97	0.05	0.95	N/A	N/A
	CF (ours)	0.06	0.98	0.09	0.91	N/A	N/A
Qwen2.5-0.5B	EAP	0.05	0.95	0.05	0.96	0.06	94.0
	EAP-IG-inputs	0.01	1.00	0.05	0.99	0.05	95.0
	CF (ours)	0.04	0.96	0.06	0.94	0.09	92.0
Llama3.2-1B	EAP	0.02	0.99	0.04	1.00	0.13	0.87
	EAP-IG-inputs	0.01	0.99	0.03	0.98	0.05	95.0
	CF (ours)	0.02	0.99	0.05	0.96	0.13	0.87
OPT-1.3B	EAP	0.01	0.99	0.01	0.99	0.05	0.95
	EAP-IG-inputs	0.00	1.50	0.01	1.00	0.04	0.96
	CF (ours)	0.01	0.99	0.05	0.95	0.07	0.93

6. Experiments

6.1. Datasets and Models

6.1.1. Circuit Tracing

Benchmarks. We evaluate our method on three established circuit discovery tasks: Indirect Object Identification (IOI) (Wang et al., 2023), Subject-Verb Agreement (SVA), and Multiple-Choice Question Answering (MCQA). For IOI and MCQA, we use the standardized prompts from the Mechanistic Interpretability Benchmark (MIB) (Mueller et al., 2025), while we generate our own prompts for the country-capital task.

Models. We test across four model families of varying sizes: GPT-2 Small (Radford et al., 2019), Qwen2.5-0.5B (Yang et al., 2024), Llama 3.2-1B (Grattafiori et al., 2024), and OPT-1.3B (Zhang et al., 2022). This selection spans different architectural choices and parameter counts, enabling us to assess the generalizability of our approach.

Evaluation. We evaluate circuit quality using the two metrics proposed by Mueller et al. (2025): the integrated Circuit Performance Ratio (CPR) and the integrated Circuit-Model Distance (CMD). CPR measures the area under the faithfulness curve, capturing how well the discovered circuit preserves task performance. CMD quantifies the area between the faithfulness curve and optimal performance, measuring how closely the circuit approximates the full model's behavior.

We compare against Edge Attribution Patching (EAP) and EAP-IG-inputs (Syed et al., 2023; Nanda, 2023), one of the leading gradient-based circuit discovery methods. Our goal is not to outperform these methods, but to demonstrate that geometric structure in answer tokens captures the necessary circuit information, supporting our claim that circuits are geometrically encoded in token representations.

6.1.2. Steering Vectors

We validate the writing aspect through two experiments. First, we test whether patching the circuits discovered via geometric tracing—using the same answer token directions—causally affects model outputs as predicted. Second, we test whether steering along these directions can redirect generation toward target outputs. Success in both would confirm that circuit fingerprints support not just discovery but intervention.

For language steering, we use the 100 English prompts from Konen et al. (2024), divided into factual and subjective categories, and steer model outputs toward five different emotions: joy, anger, sadness, surprise and disgust in Llama3.2-1B.

6.2. Circuit Tracing Results

Beyond edge-level attribution, our method also provides Shapley decompositions that quantify the relative contributions of query, key, and value computations within each attention head. From Wang et al. (2022), we know that 15 attention heads are critical to the IOI circuit in GPT2-Small. In Fig. 4, we plot a ternary diagram showing the Q/K/V importance ratios at the final token position across all attention heads. We further divide heads by the sign of their total contribution: negative heads include Duplicate Token Heads, Negative Name Movers, and Previous Token Heads, while positive heads include S-Inhibition Heads, Name Movers, and Induction Heads. This sign-based clustering aligns with the expected functional roles—negative heads suppress incorrect completions, while positive heads promote the correct indirect object.

As shown in Table 1 and Fig.3, our method achieves comparable performance to gradient-based baselines (EAP, EAP-IG) using only geometric alignment with answer token di-